

# SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

## ASSESSMENT TOOL 3 – Technical Presentation

|                  |                                                                                                               |               |                                            |
|------------------|---------------------------------------------------------------------------------------------------------------|---------------|--------------------------------------------|
| Course Code:     | CSA0309                                                                                                       | Course Title: | Data Structures                            |
| CO Assessed:     | CO1 – Imitation (BL3, BL4)                                                                                    | Assessment:   | Assessment Tool 3 – Technical Presentation |
| Weightage:       | 35% of CIA                                                                                                    | Total Marks:  | 100                                        |
| CO1 Statement:   | Apply suitable data structures and ADTs for efficient problem solving by analyzing time and space complexity. |               |                                            |
| Student Name:    | Bhuvaneshwaran J                                                                                              | Reg. No.:     | 192511402                                  |
| Section / Batch: | Slot B                                                                                                        | Date:         | 13/07/2026                                 |

### Code of Conduct

I Bhuvaneshwaran J(192511402) (Name / Reg No)

certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Bhuvaneshwaran J

### Instructions

- This assessment contains technical presentation. Total: 100 Marks.
- When a student delivers a technical presentation on a data structures topic, evaluation should be based on the following requirements: Concept explanation, Real-world application and Clarity of presentation.
- Demonstrate clear understanding, not just reading slides
- Use of tools: PowerPoint / Google Slides.
- Duration: 7–10 minutes presentation + 3 minutes Q&A.

| Section / Topic | Focus                  | Q Nos | Marks     |
|-----------------|------------------------|-------|-----------|
| Data Structure  | Real-Time Applications | 1     | 100       |
| GRAND TOTAL:    |                        |       | 100 Marks |

## Annexure A – Technical Presentation

A web browser supports navigating backward through previously visited pages. Recommend a suitable ADT and prepare a technical presentation explaining how it improves performance efficiency. (100 Marks)

(OR)

A music streaming application maintains a playlist where songs can be added, removed, or reordered dynamically. Recommend a suitable data structure and justify based on flexibility and efficiency. Prepare a technical Presentation. (100 Marks)

(OR)

A metro rail network maps stations and routes to determine reachable stations from a source station. Suggest a suitable data structure for representation and justify your answer. (100 Marks)

(OR)

A calculator application evaluates arithmetic expressions entered by users. Recommend a suitable data structure and justify based on expression processing efficiency. (100 Marks)

(OR)

A cafeteria token system serves customers strictly in the order of arrival. Identify the suitable ADT and justify why it ensures fairness. (100 Marks)

### Rubrics for Calculation

| Criteria              | Weightage | Level 4 – Exemplary                                                                          | Level 3 – Proficient                             | Level 2 – Developing                               | Level 1 – Beginning                         |
|-----------------------|-----------|----------------------------------------------------------------------------------------------|--------------------------------------------------|----------------------------------------------------|---------------------------------------------|
| Concept Understanding | 20        | Demonstrates deep and accurate understanding of data structure with insights and connections | Shows clear understanding with minor gaps        | Basic understanding; some misconceptions present   | Limited or incorrect understanding          |
| Content Organization  | 30        | Logical, well-structured, smooth flow; strong introduction and conclusion                    | Mostly organized with clear flow                 | Some organization but lacks clarity or transitions | Disorganized, difficult to follow           |
| Technical Depth       | 20        | Includes algorithms, complexity analysis, and advanced insights                              | Includes algorithm and basic complexity analysis | Limited technical details; missing depth           | Minimal or no technical content             |
| PowerPoint Slides     | 20        | Excellent design, clear visuals, enhances understanding                                      | Good slides with minor issues                    | Basic slides; somewhat cluttered or unclear        | Poor slides or no visual support            |
| Communication Skills  | 10        | Clear, confident, engaging delivery; excellent articulation                                  | Clear delivery with minor hesitation             | Some hesitation; unclear in parts                  | Poor communication; difficult to understand |

| Faculty In-charge | Course Coordinator | Program Director |
|-------------------|--------------------|------------------|
| Signature: _____  | Signature: _____   | Signature: _____ |
| Date: _____       | Date: _____        | Date: _____      |



**SIMATS**  
ENGINEERING



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Saveetha Institute of Medical And Technical Sciences  
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

# Metro Rail Network Representation Using Graph Data Structure

PRESENTED BY  
BHUVANESHWARAN J  
192511402

UNDER THE GUIDANCE:  
Dr. Uma Priyadarshini

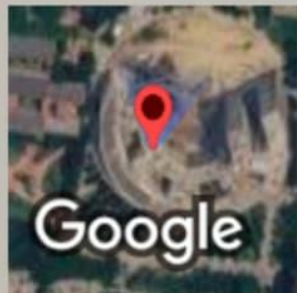


## INTRODUCTION



### What is a Graph Data Structure?

- A **graph** is a non-linear data structure.
- It consists of **Vertices (Nodes)** and **Edges (Connections)**.
- In a metro network:
  - **Stations** are represented as **Vertices**.
  - **Rail tracks** connecting stations are represented as **Edges**.
- Graphs efficiently model transportation networks.



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